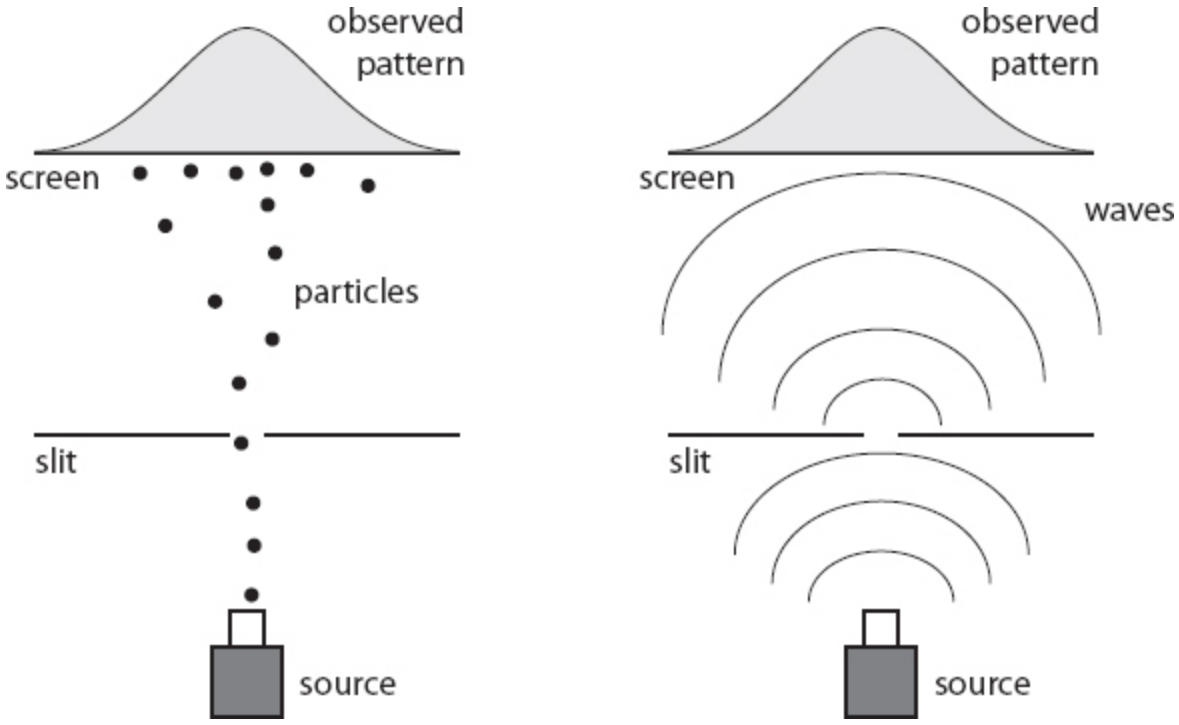


So far, that's the classical world we know and love, where particles and waves are different things and everyone can easily distinguish between them. Now let's replace our pellet gun or wave machine with a source of electrons, in all their quantum-mechanical glory. There are several twists on this setup, each with provocative consequences.

First consider just a single slit. In this case the electrons behave just as if they were classical particles. They pass through the slit, then are detected by the screen on the other side, each electron leaving a single particle-like mark. If we let numerous electrons through, their marks are scattered around a central line in the image of the slit that they passed through. Nothing funny yet.